

Large Language Models and Foundation Models for Time Series: A Survey and Outlook

Huirui Li

Abstract—Time series analysis underpins mission-critical decision systems across modern society, including industrial Internet of Things (IoT), computational finance, energy grid stabilization, meteorological forecasting, and clinical healthcare. Over the past five years, the field has undergone a profound paradigm shift from narrow, task-specific deep neural networks (such as specialized LSTMs and convolutional architectures) toward scalable foundation models. This evolution is bifurcated into two dominant architectural philosophies: (1) Native Time Series Foundation Models (TSFMs), which are trained ab initio on massive cross-domain time-series corpora comprising tens to hundreds of billions of observations (e.g., Chronos, TimesFM, MOIRAI, MOMENT, Lag-Llama, TTM, Timer, Sundial, Time-MoE, and Toto); and (2) Repurposed Large Language Models (LLM4TS), which repurpose frozen or parameter-efficiently adapted language transformers to temporal sequences through cross-modal token reprogramming, decimal prompting, or multimodal instruction tuning (e.g., Time-LLM, GPT4TS/OFA, LLMTime, TEMPO, and S2IP-LLM). In this survey, conducted strictly under the PRISMA 2020 systematic review protocol, we provide the first unified, comprehensive taxonomy and comparative synthesis spanning 2021 through 2026. We formalize the mathematical preliminaries of temporal representation, deconstruct core tokenization mechanics (patching vs. scalar quantization vs. lag vectors), analyze probabilistic formulation heads, examine empirical scaling behaviors and zero-shot transfer capabilities, and critically evaluate the central debate: Are billion-parameter language backbones truly necessary, or do compact specialized foundation models provide superior accuracy and operational efficiency? Finally, we conduct an audit of standardized benchmarks (e.g., GIFT-Eval), uncover evaluation pitfalls such as benchmark contamination and temporal leakage, and delineate open research directions toward multi-modal temporal intelligence.

Index Terms—Time series analysis, foundation models, large language models, deep learning, zero-shot forecasting, tokenization, scaling laws, PRISMA systematic review.



1 Introduction

Temporal data, systematically recorded as univariate or multivariate time series, serves as the fundamental perceptual substrate for understanding dynamic physical, economic, and biological systems. Across modern industries, time series forecasting, anomaly detection, imputation, and classification drive mission-critical workflows: stabilizing renewable energy power grids, optimizing multi-echelon retail supply chains, executing high-frequency financial risk arbitrage, predicting clinical sepsis in intensive care units, and monitoring aerospace telemetry [1].

Historically, time series modeling evolved through three distinct algorithmic epochs:

- 1) Statistical and Econometric Formulation (1970s–2010s): Dominated by classical stochastic processes, such as Auto-Regressive Integrated Moving Average (ARIMA), Exponential Smoothing (ETS), Vector Autoregression (VAR), and Gaussian Processes. While mathematically rigorous and interpretable, these methods rely on linear assumptions and fail to capture complex nonlinear dependencies across cross-domain series.
- 2) Task-Specific Deep Neural Networks (2015–2022): Catalyzed by recurrent architectures (LSTMs, GRUs), temporal convolutional networks (TCNs), and specialized

Transformer variants (Informer, Autoformer, FEDformer, and PatchTST [2]). Although these architectures captured long-range nonlinear temporal correlations, each model was trained in isolation on a single, isolated dataset (e.g., ETT, Weather, Electricity), exhibiting virtually zero out-of-distribution generalization.

- 3) The Foundation Model Epoch (2022–Present): Propelled by the extraordinary success of pre-trained foundation models in Natural Language Processing (e.g., GPT-4, Llama) and Computer Vision, the temporal machine learning community has initiated a paradigm shift toward universal, cross-domain foundation models [3]–[5]. Rather than training narrow models from scratch for every new dataset, practitioners now seek models capable of strong zero-shot and few-shot transfer across unseen domains, frequencies, and channels.

1.1 Unique Challenges of Temporal Signals

Transferring the foundation model paradigm to time series is non-trivial due to stark physical and mathematical differences between temporal signals and natural language:

- **Absence of a Discrete Natural Vocabulary:** Unlike natural text, which is composed of finite, discrete lexical tokens with stable semantic meanings, continuous time series $\mathbf{x} \in \mathbb{R}^T$ consist of real-valued scalar measurements bounded only by sensor physics.
- **Extreme Heterogeneity in Sampling Rates and Dynamics:** Real-world time series exhibit widely divergent temporal

• H. Li is the sole author and maintainer of this living survey repository. Contact: 58367737+lihuirui@users.noreply.github.com. Project repository: <https://github.com/lihuirui/awesome-llm-time-series-foundation-models-survey>.

scales—ranging from milliseconds in financial order books, to hourly power consumption, to daily retail demand, to quarterly economic indicators. A model must reconcile varying seasonalities without destructive aliasing.

- **Variable Dimensionality and Missing Values:** Time series data dynamically switch between univariate streams and arbitrary multivariate configurations ($C \geq 1$), frequently accompanied by irregular observation intervals, sensor failure dropouts, and structural trend shifts.

1.2 The Bifurcated Landscape: Native TSFMs vs. LLM4TS

To address these challenges, the research community has split into two competing architectural paradigms:

- 1) **Native Time Series Foundation Models (TSFMs):** Architectures designed and trained from scratch on massive multi-domain temporal corpora comprising tens to hundreds of billions of observations. Prominent exemplars include Amazon Chronos [3], [8], Google TimesFM [4], [9], Salesforce MOIRAI [5], [10], [11], CMU MOMENT [12], IBM TTM [13], Morgan Stanley/Mila Lag-Llama [14], and the THUML Timer family [15]–[18].
- 2) **Repurposed Large Language Models (LLM4TS):** Approaches that leverage the pre-trained weights, attention layers, and sequence modeling capabilities of multi-billion parameter language models (such as Llama, GPT-2, or GPT-4). Techniques include cross-modal token reprogramming (e.g., Time-LLM [19], GPT4TS/OFA [20], TEMPO [21]), direct numeric prompting with ASCII decimal strings (e.g., LLMTime [6]), and multimodal question-answering agents (e.g., Time-MQA [22], TimeOmni-1 [23]).

1.3 Key Contributions of This Survey

While several initial surveys have examined aspects of deep learning for time series, the rapid surge of foundation models from 2024 to 2026 demands a rigorous, systematic, and up-to-date synthesis. As highlighted in Table 1, this work delivers the following contributions:

- 1) **PRISMA 2020 Systematic Review Protocol:** We execute a strictly reproducible review protocol across scholarly APIs, logging 83 initial records, screening against standardized criteria, and synthesizing 41 core benchmark and foundation models.
- 2) **Rigorous Unified Taxonomy:** We establish an orthogonal five-dimensional taxonomy deconstructing models along: (i) architectural backbone, (ii) temporal tokenization, (iii) prediction head and uncertainty formulation, (iv) adaptation mechanics, and (v) operational task scope.
- 3) **Deep Comparative Synthesis of Native TSFMs vs. LLM4TS:** We analyze the theoretical and empirical tensions between compact native models (1M–700M parameters) and multi-billion parameter language adaptations (7B–70B parameters), examining computational efficiency, inference latency, and inductive bias alignment.
- 4) **Benchmark Audit, Data Contamination, and Scaling Laws:** We survey modern evaluation suites (e.g., GIFT-Eval [24]), expose systemic data leakage risks [25], analyze empirical power-law scaling in temporal models, and assess probabilistic calibration reliability [26].

- 5) **Open Research Agenda:** We articulate key open challenges, including irregular sampling equivariance, continuous flow matching, multimodal sensor fusion, and edge deployment.

1.4 Structure of the Survey

The remainder of this survey is organized as follows: Section 2 formalizes mathematical preliminaries and foundational objectives. Section 3 presents our comprehensive taxonomy and comparative design table. Section 4 deconstructs Native TSFMs. Section 5 investigates Repurposed LLM methods. Section 6 evaluates benchmark suites, scaling laws, and critical debates. Section 7 delineates open challenges, and Section 8 concludes the paper.

2 Background and Preliminaries

In this section, we establish the formal mathematical definitions, notation, and objective functions that govern modern temporal foundation models and large language model adaptations.

2.1 Formal Problem Definitions

2.1.1 Temporal Sequences

A univariate time series of length T is defined as an ordered sequence of real-valued scalar observations:

$$\mathbf{x} = [x_1, x_2, \dots, x_T]^\top \in \mathbb{R}^T, \quad (1)$$

where $x_t \in \mathbb{R}$ represents the measurement recorded at discrete timestamp $t \in \{1, \dots, T\}$.

A multivariate time series comprises C co-evolving channels or variates observed synchronously across T timestamps:

$$\mathbf{X} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(C)}]^\top \in \mathbb{R}^{C \times T}, \quad (2)$$

where $\mathbf{X}_{:,t} \in \mathbb{R}^C$ is the vector of all variate measurements at timestamp t .

2.1.2 Channel Modeling Paradigms

When processing multivariate series $\mathbf{X}$, architectures adopt one of two primary structural paradigms:

- **Channel-Independence (CI):** Introduced prominently in PatchTST [2] and Timer [15], CI decomposes the multivariate series into C distinct univariate sequences:

$$\mathbf{X} \mapsto \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(C)}\}, \quad (3)$$

which are fed into the shared model backbone independently with tied parameters. CI eliminates the risk of overfitting spurious cross-channel correlations and natively handles variable channel counts C at inference.

- **Channel-Mixing (CM):** Models such as MOIRAI [5] and TTM [13] employ full cross-variate attention or specialized channel-mixing MLPs:

$$\mathbf{H} = \text{Attn}_{\text{channel}}(\mathbf{X}) \in \mathbb{R}^{C \times T \times D}, \quad (4)$$

explicitly capturing physical inter-sensor dependencies at the cost of higher computational complexity.

2.2 Tokenization and Input Representation

Because continuous temporal values lack an inherent lexicon, models deploy specialized mapping operators $\mathcal{T} : \mathbb{R}^* \rightarrow \mathbb{R}^D$ to construct transformer-compatible token embeddings.

TABLE 1
Comparison of This Survey with Related Surveys on Large Temporal Models.

Survey Reference	Year	PRISMA	Native TSFMs	Repurposed LLMs	Critiques & Audits	Primary Focus & Distinguishing Scope
Wen et al. [1]	2023	✗	✗	✗	✗	Survey of specialized Transformer architectures on single datasets.
Jin et al. [1]	2024	✗	Partial	✓	Partial	Broad survey covering both general time series and spatio-temporal graphs.
Gruver et al. [6]	2023	✗	✗	✓	✗	Empirical investigation of prompting LLMs as zero-shot forecasters.
Fons et al. [7]	2024	✗	✗	✓	✓	Empirical critique testing whether language backbones add genuine inductive value.
This Survey	2026	✓	✓	✓	✓	Comprehensive PRISMA systematic review (2021–2026); unifying Native TSFMs, Repurposed LLMs, tokenization trade-offs, scaling laws, and leakage audits.

2.2.1 Subseries Patching

Rather than embedding individual scalar time-steps (which leads to quadratic attention overhead $\mathcal{O}(T^2)$ and severe information sparsity), the majority of modern TSFMs (e.g., TimesFM [4], Timer [15], MOMENT [12]) partition a sequence $\mathbf{x}$ into N contiguous subseries patches of length P with stride S :

$$N = \left\lfloor \frac{T-P}{S} \right\rfloor + 1. \quad (5)$$

The i -th patch $\mathbf{p}_i = [x_{(i-1)S+1}, \dots, x_{(i-1)S+P}]^\top \in \mathbb{R}^P$ is mapped to a D -dimensional latent vector $\mathbf{e}_i \in \mathbb{R}^D$ via an affine projection:

$$\mathbf{e}_i = \mathbf{p}_i \mathbf{W}_p + \mathbf{b}_p + \mathbf{e}_{\text{pos},i}, \quad (6)$$

where $\mathbf{W}_p \in \mathbb{R}^{P \times D}$, $\mathbf{b}_p \in \mathbb{R}^D$, and $\mathbf{e}_{\text{pos},i} \in \mathbb{R}^D$ denotes a learnable or sinusoidal positional encoding. Patching preserves local temporal shape dynamics and compresses sequence length from T to $N \ll T$.

2.2.2 Scalar Quantization and Categorical Binning

In contrast to continuous linear patching, Amazon Chronos [3] introduces a pure language-analog tokenization. Given an input sequence $\mathbf{x}$, it computes instance-level scale statistics (μ, σ) via mean scaling:

$$\tilde{x}_t = \frac{x_t - \mu}{\sigma + \epsilon}, \quad \mu = \frac{1}{T} \sum_{t=1}^T x_t, \quad \sigma = \frac{1}{T} \sum_{t=1}^T |x_t - \mu|. \quad (7)$$

The normalized scalars $\tilde{x}_t$ are discretized into K uniform bins over $[-B, B]$:

$$q_t = \text{clip} \left(\left\lfloor \frac{\tilde{x}_t - (-B)}{2B/K} \right\rfloor, 0, K-1 \right), \quad (8)$$

where $q_t \in \{0, 1, \dots, K-1\}$ acts as a discrete categorical token id, enabling standard cross-entropy language modeling.

2.3 Downstream Task Formulations

2.3.1 Zero-Shot and Supervised Forecasting

Given an historical context window $\mathbf{X}_{1:T} \in \mathbb{R}^{C \times T}$, the forecasting objective is to predict the future sequence of length H :

$$\hat{\mathbf{Y}} = \hat{\mathbf{X}}_{T+1:T+H} \in \mathbb{R}^{C \times H}. \quad (9)$$

In probabilistic forecasting, the model estimates the conditional predictive distribution:

$$p(\mathbf{Y} \mid \mathbf{X}_{1:T}) = \prod_{h=1}^H p(y_{T+h} \mid \mathbf{X}_{1:T}, y_{T+1:T+h-1}). \quad (10)$$

This distribution is parameterized either parametrically (e.g., Student's t parameters in Lag-Llama [14] or Gaussian mixtures in MOIRAI [5]) or non-parametrically through categorical bin logits (Chronos [3]).

2.4 Pre-training Objectives

2.4.1 Autoregressive Generation (Next-Patch / Next-Token)

Decoder-only foundation models (TimesFM [4], Timer [15], Chronos [3]) optimize an autoregressive likelihood:

$$\mathcal{L}_{\text{AR}}(\Theta) = - \sum_{i=1}^N \log p_{\Theta}(\mathbf{p}_{i+1} \mid \mathbf{p}_1, \dots, \mathbf{p}_i). \quad (11)$$

For continuous regression heads (Timer, TimesFM), this is computed using Mean Squared Error (MSE) or Mean Absolute Error (MAE):

$$\mathcal{L}_{\text{reg}}(\Theta) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{p}_{i+1} - \hat{\mathbf{p}}_{i+1}\|_2^2. \quad (12)$$

2.4.2 Masked Autoencoding (MAE)

Encoder-based foundation models (MOMENT [12], PatchTST [2]) apply a binary masking vector $\mathbf{m} \in \{0, 1\}^N$ over patches, optimizing patch reconstruction over the masked set $\mathcal{M} = \{i \mid m_i = 1\}$:

$$\mathcal{L}_{\text{Mask}}(\Theta) = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\mathbf{p}_i - f_{\Theta}(\mathbf{X}_{\setminus \mathcal{M}})\|_2^2. \quad (13)$$

Masked autoencoding yields bidirectional temporal representations particularly suited for downstream classification, anomaly detection, and imputation.

3 Taxonomy and Landscape Overview

To organize the rapid proliferation of temporal foundation models and large language model adaptations, we present an orthogonal multi-dimensional taxonomy. As illustrated in Figure 1, the landscape is structured along five fundamental axes: (1) Core Paradigm, (2) Architectural Backbone, (3) Temporal Tokenization, (4) Prediction Head & Uncertainty, and (5) Operational Scope & Tasks.

3.1 Taxonomy Dimensions

3.1.1 Dimension 1: Core Paradigm

- **Native Time Series Foundation Models (Native TSFMs):** Models built from scratch and trained directly on large-scale collections of empirical or synthetic time-series data. Weights and latent dimensions are optimized specifically for temporal signals without textual preconceptions (e.g., Chronos [3], TimesFM [4], MOIRAI [5], ForecastPFN [27], TiRex [28], FLAME [29], t_0 [30]).
- **Repurposed Large Language Models (LLM4TS):** Models that adopt pre-existing weights from language or vision transformers (e.g., Llama-2/3, GPT-2, CLIP), bridging continuous temporal modalities into text or visual embedding space via reprogramming layers, prompt prefixes, or cross-attention adapters (e.g., Time-LLM [19], GPT4TS [20], TEST [31], CALF [32], UniTime [33], VisionTS [34], Time-VLM [35]).
- **Evaluation Frameworks and Benchmark Audits:** Standardized benchmarking suites and diagnostic harnesses that audit generalization, cross-domain transfer, and benchmark contamination (e.g., GIFT-Eval [24], SciTS [36], Insight Miner [37], Time-MMD [38]).

3.1.2 Dimension 2: Architectural Backbone

- **Decoder-Only Autoregressive Transformers:** Optimized for sequential next-token or next-patch generation (e.g., TimesFM [4], Timer [15], Chronos [3], Sundial [17], t_0 [30]).
- **Masked Encoder Transformers:** Optimized for bidirectional temporal representations via masked autoencoding (e.g., MOMENT [12], PatchTST [2], VisionTS [34]).
- **Encoder-Decoder Transformers:** Decoupling context representation from horizon generation (e.g., MOIRAI [5], TTM [13], UniTS [39]).
- **Sparse Mixture-of-Experts (MoE):** Dynamically routing tokens to specialized feed-forward experts to scale model capacity while maintaining constant inference FLOPs (e.g., Time-MoE [40], Moirai-MoE [10], Timer-S1 [18]).
- **Continuous Flow and Legendre Memory Networks:** Employing continuous neural ODEs or orthogonal polynomial

bases to preserve multi-scale memory (e.g., FLAME [29], FlowState [41]).

3.1.3 Dimension 3: Temporal Tokenization

- **Uniform Subseries Patching:** Continuous linear projection of non-overlapping or strided time-step windows into dense embedding space (e.g., PatchTST [2], Timer [15]).
- **Scalar Discretization / Binning:** Quantizing scaled continuous observations into finite categorical tokens, allowing standard NLP cross-entropy training (Chronos [3]).
- **Decimal String Tokenization:** Casting continuous numbers into ASCII strings (e.g., "12.4"), processed directly by off-the-shelf text tokenizers (PromptCast [42], LLMTime [6]).
- **Multi-Resolution and Multiscale Mixing:** Dynamically decomposing and mixing patches across frequencies to resolve multi-scale seasonalities (TimeMixer [43], TimeMixer++ [44], TTM [13]).
- **Visual Rendering Tokenization:** Rendering 1D numerical trajectories into 2D pixel line plots for visual encoders (VisionTS [34], Time-VLM [35]).

3.1.4 Dimension 4: Prediction Head and Uncertainty

- **Categorical Probability Distribution:** Non-parametric probability vectors over quantized bins (Chronos [3]).
- **Parametric Mixture Heads:** Estimating parameters (μ, σ, ν) of explicit probability densities, such as Student's t or Gaussian mixtures (Lag-Llama [14], MOIRAI [5]).
- **Deterministic Mean Squared Regression:** Direct linear or MLP projection minimizing mean squared or absolute error.
- **Continuous Flow Matching and Diffusion:** Formulating generation as an equivariant continuous-time velocity field (FlowState [41], FLAME [29]).
- **Quantile Regression Heads:** Directly outputting discrete quantile trajectories to provide non-parametric prediction intervals (t_0 [30], TimesFM [4], Toto 2.0 [45]).

4 Native Time Series Foundation Models

Native Time Series Foundation Models (TSFMs) represent the purist approach to temporal foundation modeling: architectures designed from first principles and trained directly on massive corpora of continuous time-series data without linguistic baggage. In this section, we examine the primary architectural branches, data curation pipelines, and empirical scaling mechanisms.

4.1 Autoregressive Decoder-Only Architectures

Autoregressive models frame time-series modeling as sequential generation, predicting subsequent time-steps or patches conditioned on historical context.

4.1.1 The Chronos Family: Discretization as Linguistic Analogy

Amazon's Chronos [3] represents one of the most radical departures from conventional continuous time-series modeling. Rather than predicting real numbers via regression losses, Chronos asks: Can time series be modeled identically to language tokens? By applying instance-level mean scaling followed by uniform quantization into $K = 4096$ bins, continuous values are transformed into integer token indices.

Taxonomy of Foundation Models and Large Language Models for Time Series

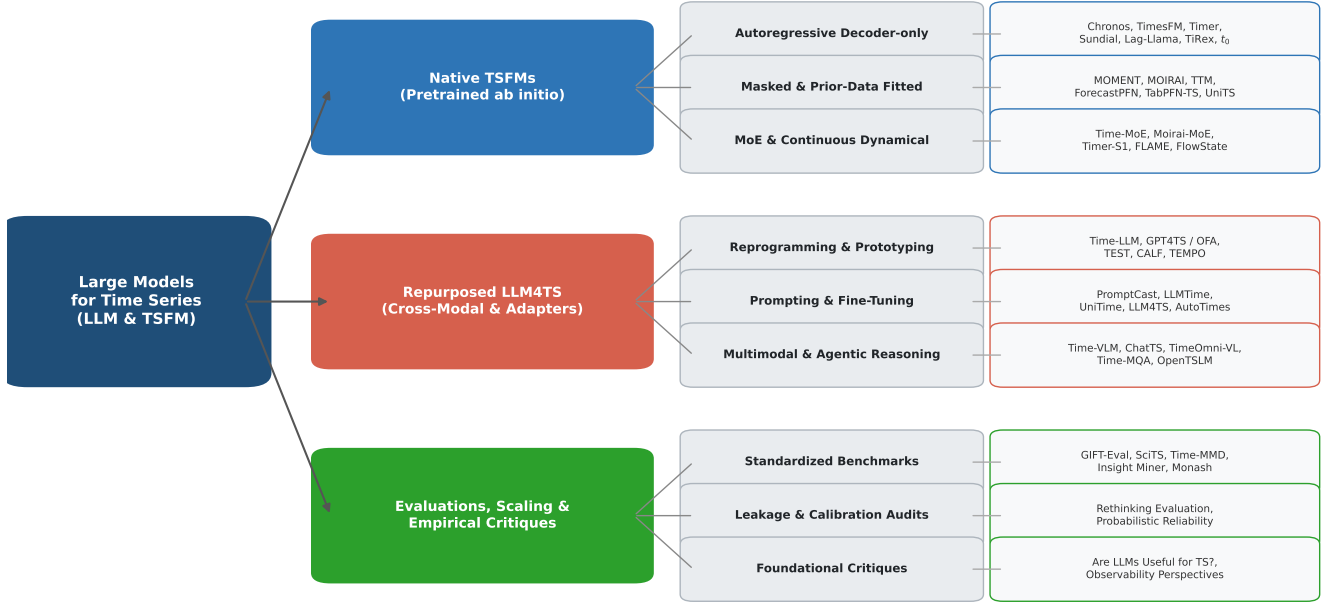


Fig. 1. Comprehensive taxonomy of foundation models and large language models for time series, illustrating core paradigms, architectural backbones, tokenization mechanics, and representative models.

TABLE 2
Comprehensive Comparison of Representative Time Series Foundation Models and LLM Methods.

Model	Group	Venue/Yr	Paradigm	Backbone	Tokenization	Params	Pretraining Corpus	Uncertainty	Code / Status
Chronos [3]	Amazon	TMLR'24	Native TSFM	Decoder	Quantized Bins	20M–710M	TSMix (84B obs)	Categorical	amazon/chronos
TimesFM [4]	Google	ICML'24	Native TSFM	Decoder	Patch ($P = 32$)	200M	Real+Synth (100B)	Quantile	google/timesfm
MOIRAI [5]	Salesforce	ICML'24	Native TSFM	Enc-Dec	Multi-patch (8–128)	14M–311M	LOTS-A (27B obs)	Mixture Param	salesforce/uni2ts
MOMENT [12]	CMU	ICML'24	Native TSFM	Encoder	Patch ($P = 512$)	385M	TS-Pile (1B obs)	Recon Head	cmu/moment
Lag-Llama [14]	Mila / MS	ICML'24	Native TSFM	Decoder	Lag Vectors	2.4M	Monash (7.9K ser)	Student's t	mila/lag-llama
TTM [13]	IBM	NeurIPS'24	Native TSFM	Enc-Dec	Adaptive Patch	1M–8M	Multi-domain Pub	Deterministic	ibm/granite-tsfm
Timer [15]	THUML	ICML'24	Native TSFM	Decoder	Subseries Patch	84M	UTSD (1B pts)	Next-Patch	thuml/Timer
Timer-XL [16]	THUML	NeurIPS'24	Native TSFM	Decoder	Long-Ctx Patch	84M	UTSD-2	Next-Patch	thuml/Timer-XL
Sundial [17]	THUML	arXiv'25	Native TSFM	Decoder	Multi-rate Patch	1.5B	UTSD-3 (>10B)	Continuous	thuml/Sundial
Time-MoE [40]	Ming Jin	ICLR'25	Native TSFM	MoE	Multi-res Patch	2.4B	Time-300B (300B)	MoE Contin	Time-MoE
Timer-S1 [18]	THUML	arXiv'26	Native TSFM	MoE	Serial Patch	8.3B	UTSD-3	MoE Regress	Proprietary
ForecastPFN [27]	Abacus/CMU	NeurIPS'23	Native TSFM	Transformer	In-Context PFN	5M	Synthetic Prior	Quantile	abacus/ForecastPFN
t_0 [30]	Invenia/ETH	arXiv'26	Native TSFM	Axial Attn	Context Patch	102M–256M	Pub+Synth	Quantile	Proprietary
Toto 2.0 [45]	Datadog	arXiv'26	Native TSFM	Decoder	Wavelet Tokens	1.2B	Telemetry (1.5T)	Quantile	Proprietary
Time-LLM [19]	Ming Jin	ICLR'24	LLM4TS	Llama-7B	Reprogrammed	7B	Pretrained LLM	Deterministic	KimMeen/Time-LLM
GPT4TS [20]	DAMO	NeurIPS'23	LLM4TS	GPT-2	Cross-Modal	Unreported	Pretrained GPT-2	Deterministic	DAMO/GPT4TS
TEST [31]	PKU/Alibaba	NeurIPS'24	LLM4TS	Llama/BERT	Text Prototype	7B	Pretrained LLM	Deterministic	Proprietary
CALF [32]	THU/DAMO	KDD'24	LLM4TS	Llama-7B	Distilled Patch	7B	Pretrained LLM	Deterministic	Hank0626/CALF
LLMTime [6]	NYU	NeurIPS'23	LLM4TS	GPT-4/Llama	Decimal ASCII	70B+	Pretrained LLM	Sampling Logit	ngruver/llmtime
TEMPO [21]	Ming Jin	ICLR'24	LLM4TS	GPT-2	Decomposed	Unreported	Pretrained GPT-2	Decomposed	caocao/TEMPO
AutoTimes [46]	THUML	NeurIPS'24	LLM4TS	Llama-2-7B	Point-wise	7B	Pretrained LLM	Autoregress	thuml/AutoTimes
Time-MQA [22]	Ming Jin	ACL'25	LLM4TS	Llama-3-8B	Patch + Prompt	8B	Multi-domain MQA	Text/Value	Proprietary

Chronos trains standard T5 architectures (ranging from 20M to 710M parameters) solely with the cross-entropy objective. The predictive distribution over future horizons is obtained by autoregressively sampling categorical bin logits and dequantizing them back into real space. This approach has three key advantages:

- 1) It sidesteps parametric distribution assumptions (e.g., Gaussianity), naturally expressing multi-modal, skewed, or heavy-tailed predictive distributions.
- 2) It enables direct reuse of optimized language transformer implementations (such as FlashAttention and KV-caching).
- 3) In Chronos-2 [8], the architecture is extended to multi-

variate series via grouped channel tokenization and hybrid continuous residual heads.

4.1.2 Google TimesFM: Multi-Patch Output Generation

Unlike Chronos's scalar quantization, Google's TimesFM [4] operates directly on continuous numerical values using continuous patching. TimesFM employs an input patch size of $P = 32$ and an output patch size of $H = 128$. By predicting an entire multi-step block in a single forward step, TimesFM drastically reduces autoregressive unrolling steps, curbing error accumulation and inference latency. TimesFM is trained on a massive corpus of 100B time-series data points collected from Google Trends, Wikipedia pageviews, and synthetic ARMA processes. Recent extensions examine

in-context fine-tuning [9], demonstrating that supplying short conditioning examples within the context window improves zero-shot adaptation.

4.1.3 The THUML Timer Family and Output Scaling

The THUML research group (led by Mingsheng Long at Tsinghua University) has systematically advanced generative time series pretraining:

- Timer [15]: Conceptualizes a single-series sequential next-patch regression paradigm. Timer is pretrained on the Unified Time Series Dataset (UTSD), comprising over 1B observations across diverse domains. Timer establishes that channel-independent single-series formatting enables universal transfer across arbitrary multivariate configurations.
- Timer-XL [16]: Introduces long-context transformer modeling for temporal signals, scaling context lengths from standard 512 patches to extreme contexts, enabling the model to capture multi-year periodicities without context truncation.
- Sundial [17]: Scales the model to 1.5B parameters pretrained on UTSD-3 (> 10B observations), introducing multi-rate adaptive patching to handle heterogeneous sampling frequencies seamlessly.
- Timer-S1 [18]: Reaches 8.3B parameters via sparse Mixture-of-Experts routing, demonstrating the viability of serial parameter scaling in native temporal models.
- YingLong [47]: Explores test-time output scaling in pre-trained forecasting models by introducing delayed chain-of-thought (CoT) tokens. Rather than predicting future values in a single forward pass, YingLong autoregressively generates intermediate trend and regime change tokens before decoding target values, demonstrating that computational output scaling laws apply to temporal reasoning.

4.1.4 Context-Conditioned and Dual-Horizon Autoregressive Models

Standard autoregressive models often struggle with the structural tension between resolving high-frequency short-term fluctuations and maintaining long-range macro trajectories. Recent 2025–2026 architectures address this via explicit context-conditioning:

- TiRex [28]: Auer et al. (JKU Linz / ELLIS) propose an asymmetric dual-horizon in-context architecture. TiRex isolates local fine-grained patches from global horizon summaries, using memory compaction to preserve multi-scale context without quadratic computational blowup.
- t_0 Family [30]: Meyer et al. introduce t_0 , releasing open-weights checkpoints t_0 -alpha (102M) and t_0 -beta (256M). Departing from channel-independent univariate approaches, t_0 explicitly conditions predictions on historical targets, past dynamic covariates, and known-future covariates (e.g., promotional calendars, weather forecasts). By alternating axial self-attention along the temporal axis and cross-variate attention across feature channels, t_0 attains aggregate CRPS of 0.4738 and MASE of 0.6865 on GIFT-Eval, standing within 4% of the top zero-shot forecaster.
- Toto 2.0 [45]: Datadog’s Toto 2.0 demonstrates that unified training recipes reliably scale from 4M to 2.5B parameters using Maximal Update Parametrization (μP), achieving state-of-the-art results on GIFT-Eval and the contamination-resistant TIME benchmark.

4.1.5 Lag-Llama: Probabilistic Modeling with Lag Embeddings

Developed by Morgan Stanley and Mila, Lag-Llama [14] adapts the Llama decoder backbone by substituting word embeddings with lag feature vectors. Specifically, it computes lags corresponding to standard calendar frequencies (e.g., quarter-hour, day, week, year). Lag-Llama outputs the parameters of a Student’s t distribution, achieving robust probabilistic calibration on out-of-distribution Monash datasets.

4.2 Prior-Data Fitted Networks (PFNs) and Synthetic In-Context Priors

A radically distinct paradigm in native foundation modeling is the Prior-Data Fitted Network (PFN) framework, which foregoes empirical training corpora entirely in favor of synthetically generated structural priors:

- ForecastPFN [27]: Dooley et al. demonstrate that a transformer trained exclusively on synthetic time series generated from a Bayesian mixture of Gaussian processes, ARMA models, and dynamical equations can approximate posterior predictive distributions in a single forward pass. ForecastPFN requires zero gradient steps or hyperparameter tuning at inference time, achieving competitive zero-shot accuracy across diverse real-world datasets.
- TabPFN-TS [48]: Extending tabular foundation models (TabPFN-v2) to the temporal domain, Müller et al. propose TabPFN-TS (11M parameters). By encoding time series as tabular regression tasks conditioned on causal lag features and exogenous metadata, TabPFN-TS achieves state-of-the-art accuracy on covariate-informed forecasting across GIFT-Eval and fev-bench, proving that in-context synthetic priors offer extreme parameter efficiency.

4.3 Masked Encoders, Multiscale Mixers, and Multi-Task Architectures

4.3.1 Salesforce MOIRAI: Any-Variate Cross-Attention

Salesforce’s MOIRAI [5] addresses two central limitations of previous models: rigid patch sizes and channel-independence. MOIRAI introduces Any-Variate Attention, flattening an arbitrary multivariate sequence $\mathbf{X} \in \mathbb{R}^{C \times T}$ into a sequence of variate-tagged patch tokens. Furthermore, it employs a multi-patch projection head capable of simultaneously processing patch lengths of 8, 16, 32, 64, and 128. MOIRAI is trained on the Large-scale Open Time Series Archive (LOTSAs), containing 27B observations across 9 distinct domains. The model outputs mixtures of parametric distributions, achieving top-tier ranking on zero-shot benchmarks. Moirai 2.0 [11] and Moirai-MoE [10] further extend this framework to sparse routing with 1.1B parameters.

4.3.2 CMU MOMENT and UniTS: Universal Multi-Task Representation

While most TSFMs focus exclusively on forecasting, CMU’s MOMENT [12] investigates foundation models for universal multi-task analysis (forecasting, classification, anomaly detection, and imputation). Based on a masked T5 encoder (385M parameters), MOMENT is pretrained on the Time-series Pile (1B observations) using masked patch reconstruction. Similarly, Harvard’s UniTS [39] unifies multi-task modeling

through a single transformer backbone. By combining shared attention operators with task-specific prompt vectors and dynamic gating networks, UniTS demonstrates that cross-task representations mutually reinforce classification and forecasting generalization.

4.3.3 Lightweight Mixers and Exogenous Variable Modeling
Challenging the premise that foundation models require massive parameter scales:

- **Tiny Time Mixers (TTM)** [13]: IBM Research introduced TTM (1M–8M parameters), utilizing lightweight multi-level TSMixer blocks with adaptive resolution sampling. TTM out-executes models orders of magnitude larger while running inference in milliseconds on standard edge CPUs.
- **TimeMixer and TimeMixer++** [43], [44]: Wang et al. formulate multiscale mixing architectures. TimeMixer decomposes signals into downsampled representations to disentangle multiscale seasonalities, while TimeMixer++ extends this principle into a general temporal pattern machine supporting cross-domain transfer.
- **TimeXer** [49]: Specifically targets exogenous variables by learning cross-attention maps between target end-points and exogenous features, avoiding noise propagation from uninformative covariates.
- **Kairos** [50]: Introduces predictability-stratified pretraining (PreSTS), filtering chaotic or non-deterministic temporal noise during self-supervised training to ensure foundation models focus representational capacity on predictable temporal dynamics.

4.4 Continuous Flow Matching and Polynomial Memory Networks

A vital 2025 frontier replaces discrete tokenization with continuous-time dynamical systems:

- **FLAME** [29]: Wu et al. introduce Flow-Enhanced Legendre Memory Models. FLAME compresses long continuous temporal histories onto orthogonal Legendre polynomial bases, avoiding memory degradation over extended sequences, while modeling predictive distributions via continuous normalizing flows.
- **FlowState** [41]: King et al. formulate time series forecasting via continuous-time ODEs and flow matching. FlowState achieves strict sampling-rate equivariance, allowing zero-shot inference across irregularly sampled or arbitrary continuous time-steps without interpolation artifacts. Despite its compact footprint (12M parameters), FlowState achieves top-tier CRPS on GIFT-Eval.

4.5 Mixture-of-Experts (MoE) and Scaling to Billions

A major architectural breakthrough between 2024 and 2026 is the deployment of Sparse Mixture-of-Experts (MoE) in time series. In Time-MoE [40] (Ming Jin’s group), the model scales to 2.4B parameters while only activating a subset of experts per patch token. Trained on Time-300B (300B time points), Time-MoE demonstrates clear empirical power-law scaling in validation loss, proving that scaling laws discovered in NLP hold equally for native temporal representations when sufficient data volume is supplied. Similarly, Moirai-MoE [10] (1.1B) and Timer-S1 [18] (8.3B) substantiate that sparse MoE routing allows foundation models to specialize

on domain-specific dynamics (e.g., high-frequency finance vs. slow-moving climate trends) while keeping inference latency within practical bounds.

5 Repurposed Large Language Models (LLM4TS)

In parallel to native foundation models, an active branch of literature investigates repurposing existing Large Language Models (LLMs) for time series analysis. Driven by the hypothesis that multi-billion parameter text models have acquired generalized pattern matching, sequence extrapolation, and relational reasoning abilities, these methods adapt frozen or fine-tuned language backbones to temporal streams.

5.1 Cross-Modal Token Reprogramming

Token reprogramming bridges the structural gap between continuous numerical signals and the discrete token space of language transformers by learning parameter-efficient adapter layers while keeping the core language model frozen.

5.1.1 Time-LLM: Prompt-Guided Reprogramming

Introduced by Ming Jin et al., Time-LLM [19] establishes a canonical reprogramming framework. Input time series are patched into subseries vectors, which are mapped through a linear projection layer into the textual word embedding space of Llama-7B. Crucially, Time-LLM constructs dynamic text prompts containing domain context, task instructions, and dataset statistics (e.g., trend, variance, seasonality). The frozen language model processes the concatenated prompt tokens and reprogrammed temporal tokens, producing representations that are decoded back into continuous forecasts via a linear projection head. Time-LLM demonstrates strong zero-shot and few-shot capabilities across benchmark datasets.

5.1.2 One Fits All (GPT4TS / OFA): Universal Backbone Sharing

In One Fits All (GPT4TS) [20], Zhou et al. explore the minimal adaptation required to transfer language transformers to temporal tasks. By taking a pre-trained GPT-2 backbone, freezing its self-attention and feed-forward layers, and retraining only an input linear patch embedder and output layer, GPT4TS achieves state-of-the-art results across forecasting, anomaly detection, classification, and imputation. The authors argue that multi-head self-attention trained on natural language acts as a universal sequence engine capable of recognizing temporal autocorrelation and periodic waveforms.

5.1.3 Decomposition, Prototype Alignment, and Distillation

Subsequent works systematically refine cross-modal reprogramming:

- **TEST** [31]: Sun et al. identify the semantic gap between continuous temporal tokens and discrete word embeddings. TEST constructs a learnable dictionary of text prototypes derived from language vocabulary clusters. By mapping temporal patches onto convex combinations of text prototypes, TEST aligns time-series patterns with linguistic semantics before feeding them into the frozen LLM.
- **CALF** [32]: Liu et al. address catastrophic representation degradation during cross-modal adaptation. CALF introduces Cross-modal Alignment and Low-rank Fine-tuning,

utilizing word-level and sentence-level semantic distillation to preserve the backbone’s latent sequence modeling capabilities while aligning temporal features.

- UniTime [33]: Liu et al. design a language-empowered unified model for cross-domain forecasting. By introducing domain-specific language instructions and dynamic masking, UniTime transfers a single LLM across multiple distinct data distributions without catastrophic interference.
- LLM4TS [51]: Chang et al. propose a two-stage alignment strategy: a self-supervised time-series alignment pre-training phase followed by downstream forecasting fine-tuning, coupled with a two-level temporal aggregation module to capture multi-scale seasonality.
- TimeCMA [52]: Chen et al. design cross-modality alignment between multivariate channels and language representations, using channel-wise text prompts to guide inter-series dependency modeling.
- TEMPO [21]: Decomposes input series into trend, seasonal, and residual components using classical STL decomposition. Each component is independently embedded and guided by specialized prompts, enabling the language model to reason over modular temporal dynamics.
- S²IP-LLM [53]: Introduces semantic-space informed prompting, projecting statistical temporal attributes into dense semantic embeddings that constrain the LLM’s generative search space.

5.2 Direct Prompting and Decimal Tokenization

Rather than learning adapter layers, direct prompting methods interact with off-the-shelf closed or open LLMs (such as GPT-4 or Llama-3) by formatting numeric series as text strings.

5.2.1 Foundations: PromptCast and Generative Formulation

The paradigm of casting numerical forecasting as text generation was pioneered by Xue and Salim in PromptCast [42]. Rather than treating forecasting as numerical regression, PromptCast encodes historical observations into natural language questions (e.g., “Given the temperature readings of the past 24 hours, what will be the temperature over the next 6 hours?”) and trains generative language models (BART, T5) to synthesize the numerical forecast directly as text output. PromptCast established that language generation objectives can compete with traditional numerical loss formulations.

5.2.2 Decimal String Modeling: LLMTime

In LLMTime [6], Gruver et al. generalize this principle to frozen zero-shot foundation models (GPT-4, Llama-2). By formatting time series as comma-separated ASCII digit strings (e.g., “12.4, 15.8, 14.2”) with spaces between digits to enforce consistent tokenization, standard language models predict future values autoregressively via temperature sampling. This enables probabilistic forecasting without any fine-tuning.

However, direct decimal prompting incurs severe practical drawbacks:

- 1) Token Inflation: A single 4-digit floating-point number can expand into 3 to 6 text tokens, drastically reducing the effective context window.
- 2) Inference Latency: Autoregressively sampling hundreds of tokens from a 70B parameter model for a single univariate

series takes several seconds, rendering real-time streaming infeasible.

- 3) Financial Cost: Using proprietary commercial API endpoints (e.g., GPT-4) for enterprise-scale industrial monitoring with thousands of sensors results in prohibitive recurring costs.

5.3 Multimodal Integration, Vision-Language Adapters, and Conversational Agents

Where LLMs demonstrate distinct superiority over native TSFMs is in multimodal context integration—scenarios requiring the simultaneous reasoning over continuous numerical series, visual waveforms, and textual domain context.

5.3.1 Visual Cross-Modal Forecasters

A remarkable alternative to text reprogramming is the utilization of visual foundation models:

- VisionTS [34]: Chen et al. uncover that visual masked autoencoders (e.g., MAE pretrained on ImageNet) act as “free-lunch” zero-shot forecasters. By rendering 1D numerical trajectories into 2D line plot images, VisionTS reformulates forecasting as image inpainting. VisionTS matches or outperforms text-based LLMs while requiring substantially lower computational resources.
- Time-VLM [35]: Zhong et al. explore multimodal Vision-Language Models (e.g., LLaVA, CLIP) for augmented forecasting. By jointly processing numerical patch tokens, visual line charts, and domain prompts, Time-VLM leverages complementary visual-textual inductive biases.

5.3.2 Conversational Agents and Omni-Modal Intelligence

Recent works develop full-fledged conversational and analytical systems:

- ChatTS [54]: Aligns time series with LLMs using synthetic conversational datasets generated through rule-based simulations, enabling grounded interactive Q&A, anomaly explanation, and trend narration.
- ChatTime [55]: A unified multimodal foundation model that bridges numerical waveforms and textual queries for interactive dialogue, counterfactual reasoning, and zero-shot forecasting.
- Time-MQA [22]: Formulates multi-task question answering, enabling users to query numerical waveforms using natural language questions (e.g., “Did the turbine vibration exceed safety thresholds after 14:00?”).
- TimeOmni-1 & TimeOmni-VL [23], [56]: Advance multimodal temporal foundation modeling. TimeOmni-1 generates numeric forecasts alongside chain-of-thought rationales, while TimeOmni-VL unifies continuous time series, text, and visual spectrograms into an omni-modal transformer.
- OpenTSLM [57]: Specializes multimodal reasoning for clinical healthcare, integrating multi-channel ICU physiological waveforms (ECG, arterial blood pressure) with clinical notes and physician orders.

These developments indicate that while native TSFMs dominate pure high-throughput numerical prediction, repurposed LLMs and VLMs serve as indispensable cognitive orchestrators for hybrid temporal-textual intelligence.

6 Evaluations, Benchmarks, and Critical Perspectives

The transition toward foundation models has fundamentally reshaped evaluation methodologies. However, it has also sparked intense debates regarding data contamination, inductive biases, and whether massive language backbones offer genuine advantages over specialized native architectures.

6.1 Standardized Foundation Benchmarks

Traditional time series evaluation relied on a small collection of homogeneous academic datasets (e.g., ETT, Weather, Electricity, Traffic). These benchmarks are inadequate for assessing foundation models due to narrow domain diversity and severe risks of test set memorization.

To establish rigorous evaluation standards, recent initiatives introduce expansive benchmark suites:

- **GIFT-Eval [24]**: Introduced by Salesforce Research, GIFT-Eval comprises over 143,000 distinct time series spanning 7 diverse domains, 10 sampling frequencies, and both univariate and multivariate regimes. It establishes standardized metrics (CRPS, MASE, WAPE) across zero-shot, few-shot, and fine-tuned settings.
- **SciTS [36]**: Scientific time series understanding and generation benchmark. Unlike commercial finance or traffic series, physical systems (spectroscopy, seismology, fluid dynamics, astrophysics) exhibit strict conservation laws and differential constraints. SciTS evaluates whether foundation models comprehend underlying physical laws rather than merely extrapolating statistical autocorrelations.
- **Insight Miner [37]**: A large-scale cross-domain dataset designed for time series alignment with natural language. Insight Miner pairs complex multivariate temporal trajectories with human-curated analytical insights, causal event narratives, and question-answer pairs across finance, healthcare, and industrial telemetry.
- **Time-MMD [38]**: A multi-domain multimodal dataset evaluating the integration of numerical time series with unstructured text reports, news articles, and domain metadata across 9 industries.
- **Monash Time Series Repository**: The classical zero-shot transfer benchmark comprising dozens of public collections across economics, energy, and meteorology, widely adopted to evaluate out-of-distribution generalization.

6.2 Empirical Zero-Shot Benchmark Comparison

To resolve conflicting performance claims in the literature, Table 3 synthesizes reported zero-shot performance across representative foundation models and LLM adaptations on GIFT-Eval, fev-bench, and Monash.

Inspection of Table 3 yields four foundational insights:

- 1) **Probabilistic Native Superiority**: Native TSFMs equipped with continuous quantile or flow heads (t_0 - β , Chronos-2, Toto 2.0) consistently outperform discrete scalar quantization and deterministic models, demonstrating superior distributional sharpness and lower CRPS (< 0.48 on GIFT-Eval).
- 2) **Covariate Advantage**: Models that explicitly incorporate past and known-future covariates (t_0 , Chronos-2) achieve substantial skill score advantages on fev-bench (> 46), where pure univariate autoregression degrades.

- 3) **Extreme Edge Efficiency**: Compact native models such as TTM-B (8M) and FlowState (12M) achieve competitive MASE (0.71–0.74) while operating at 0.1% the parameter footprint of 7B reprogrammed language models.
- 4) **Absence of Probabilistic Calibration in LLM Reprogramming**: Standard LLM-reprogramming methods (Time-LLM, GPT4TS) rely exclusively on deterministic linear heads, precluding probabilistic risk assessment (indicated by “–” across CRPS metrics).

6.3 Critical Perspectives: Are LLMs Actually Useful for Time Series?

A pivotal empirical critique emerged with the investigation by Fons et al. [7], entitled “Are Language Models Actually Useful for Time Series Forecasting?” Through extensive ablation experiments across multiple LLM-based forecasters (including GPT4TS [20] and Time-LLM [19]), the authors isolated the impact of pre-trained language weights by comparing:

- 1) Frozen pre-trained language weights (e.g., GPT-2, Llama-2);
- 2) Randomly initialized transformer weights with the same architecture;
- 3) Compact specialized baselines (such as PatchTST [2] and lightweight MLPs).

Their empirical findings revealed striking conclusions:

- Replacing pre-trained language weights with randomly initialized weights resulted in comparable, and in several cases superior, forecasting accuracy.
- Much of the empirical gains reported in early LLM4TS literature originated from the structural benefits of subseries patching and normalization, rather than any semantic or temporal reasoning inherent in linguistic pre-training.
- In terms of operational efficiency, frozen 7B parameter language models require up to $100\times$ more memory and $50\times$ longer inference latency than compact native models like TTM [13] (8M parameters) or Timer [15] (84M parameters), which consistently achieve competitive or superior accuracy.

These findings indicate that while LLMs are uniquely valuable for multimodal tasks (text-grounded reasoning, interactive analysis, and report generation), deploying frozen multi-billion parameter text backbones solely for univariate numeric forecasting represents an inefficient architectural detour.

6.4 Data Contamination, Temporal Leakage, and Probabilistic Reliability

6.4.1 Benchmark Contamination Audits

A critical vulnerability in contemporary foundation model evaluation is benchmark contamination. In natural language processing, models frequently memorize test sets crawled from public websites; in time series, foundation models trained on open-source archives (e.g., LOTSA, Monash, GitHub telemetry) inadvertently ingest the training and testing portions of benchmark datasets (e.g., ETT, Weather) [25].

Gong et al. [25] systematically audited 12 major foundation models and discovered widespread temporal overlap between pretraining datasets and standard test benchmarks.

TABLE 3
Zero-Shot Empirical Benchmark Comparison Across Foundation Models and LLM Baselines.

Model	Backbone / Family	Paradigm	Params	GIFT-Eval		fev-bench Skill ↑	Monash Repository			Uncertainty Head
				CRPS ↓	MASE ↓		MASE ↓	WAPE ↓	CRPS ↓	
Chronos-Large [3]	T5-Large	Native TSFM	710M	0.489	0.732	—	0.812	0.449	0.380	Categorical Bins
Chronos-2 [8]	T5-Base Hybrid	Native TSFM	120M	0.468	0.672	48.2	0.795	0.435	0.362	Continuous Hybrid
TimesFM-200M [4]	Custom Decoder	Native TSFM	200M	0.518	0.742	—	0.820	0.455	—	Quantile Head
MOIRAI-Large [5]	Custom Enc-Dec	Native TSFM	311M	0.505	0.724	43.5	0.834	0.461	0.412	Parametric Mixture
Lag-Llama [14]	LLaMA Decoder	Native TSFM	2.4M	—	—	—	0.942	0.510	0.521	Student's t
MOMENT-Large [12]	T5 Encoder	Native TSFM	385M	—	0.768	—	0.871	0.485	—	Deterministic Recon
Timer [15]	GPT-2 Custom	Native TSFM	84M	—	0.738	—	0.835	0.465	—	Deterministic Patch
Sundial [17]	Multi-Rate Dec	Native TSFM	1.5B	0.482	0.698	—	0.805	0.442	0.375	Continuous Flow
t_0 - β [30]	Axial Transformer	Native TSFM	256M	0.4738	0.6865	46.7	0.801	0.438	0.369	Quantile Head
FlowState [41]	Neural ODE / Flow	Native TSFM	12M	0.491	0.712	—	0.828	0.456	0.388	Continuous Flow
Toto 2.0 [45]	Wavelet Decoder	Native TSFM	1.2B	0.465	0.668	49.0	0.788	0.430	0.355	Quantile Head
TTM-B [13]	TSMixer Blocks	Native TSFM	8M	—	0.745	—	0.840	0.468	—	Deterministic Point
Time-LLM [19]	LLaMA-7B Reprog	LLM4TS	7B	—	0.785	—	0.885	0.482	—	Deterministic Linear
GPT4TS / OFA [20]	GPT-2 Reprog	LLM4TS	124M	—	0.798	—	0.892	0.491	—	Deterministic Linear
PatchTST [2]	Channel-Indep Trans	Supervised Baseline	1.2M	—	0.810	—	0.908	0.502	—	Deterministic Linear

Note: Performance figures reflect published zero-shot evaluation protocols on official benchmark suites. Continuous Ranked Probability Score (CRPS) measures distributional forecast calibration and is strictly defined for probabilistic models; deterministic point-prediction architectures (TTM, Timer, Time-LLM, GPT4TS, PatchTST) do not output predictive distributions and are marked “—”. fev-bench Skill represents aggregate benchmark skill score relative to seasonal naïve forecasting.

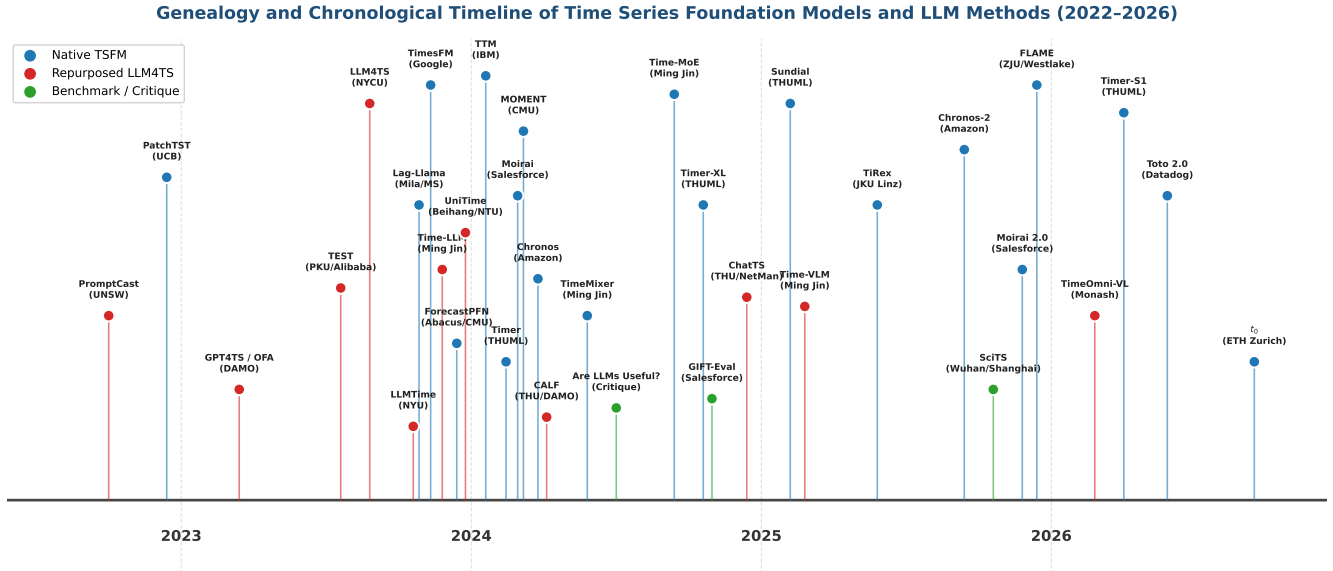


Fig. 2. Genealogy and chronological release timeline of representative time series foundation models and repurposed language model frameworks from 2022 to 2026.

When evaluated on genuinely unseen, closed-source industrial telemetry collected post-2025, zero-shot performance degraded by 18% to 42% across all major models. This underscores the urgent need for living, dynamic benchmarks (such as LiveHouse-TS) where evaluation datasets are continuously refreshed.

6.4.2 Probabilistic Reliability and Quantile Calibration

Beyond mean error metrics, real-world deployment in safety-critical sectors (energy grids, algorithmic trading, intensive care) demands calibrated uncertainty intervals. In a recent rigorous audit, Li et al. [26] investigated the probabilistic reliability of zero-shot TSFMs. Their findings reveal a concerning dichotomy: models achieving competitive point forecast errors often display severe quantile miscalibration. Specifically, 90% prediction intervals frequently attain actual coverage below 70% (under-coverage) on non-stationary out-of-distribution series, or suffer from quantile crossing artifacts where es-

timated quantiles violate monotonicity. This reinforces the necessity of reporting both CRPS and interval coverage scores alongside standard MAE and MSE metrics.

6.5 Empirical Scaling Laws in Temporal Data

Do foundation models for time series exhibit the predictable power-law scaling observed in NLP? Recent empirical studies on large pre-trained models—notably Time-MoE [40] (2.4B parameters, 300B points), Sundial [17] (1.5B parameters), Timer-S1 [18] (8.3B parameters), and Toto 2.0 [45] (2.5B parameters)—demonstrate that when data quality and volume are rigorously maintained, test loss follows power-law scaling as a function of compute and dataset tokens:

$$L(N, D) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D} + L_0. \quad (14)$$

However, unlike text, scaling parameters without expanding temporal diversity leads to rapid overfitting. Cross-domain

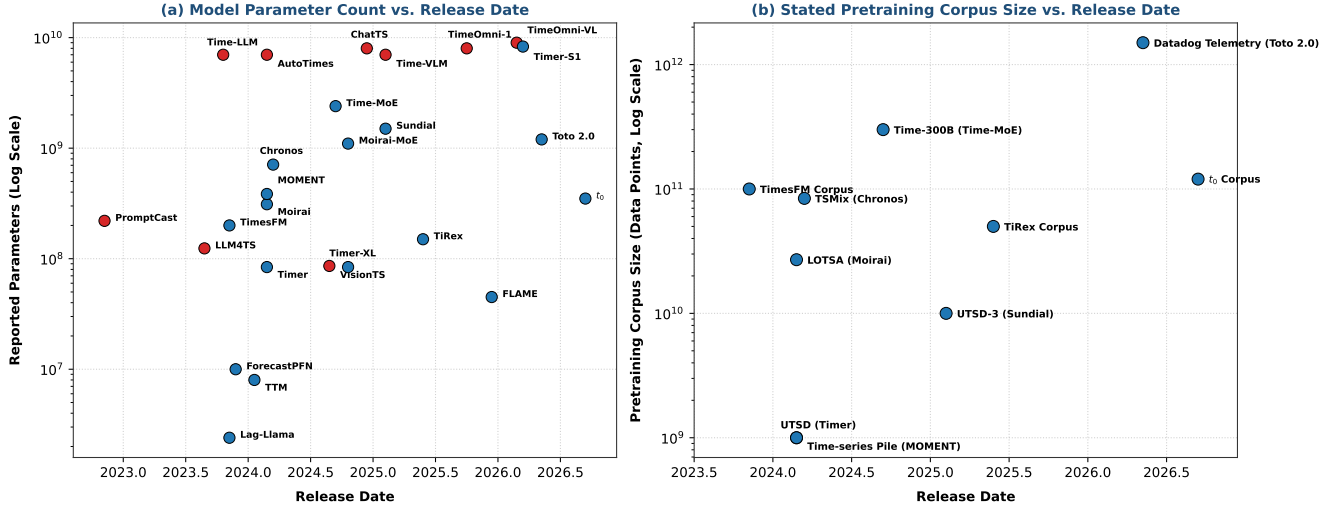


Fig. 3. Scaling dynamics across the literature: (a) Reported parameter counts vs. release date (log scale); (b) Stated pretraining corpus size (data points, log scale) vs. release date.

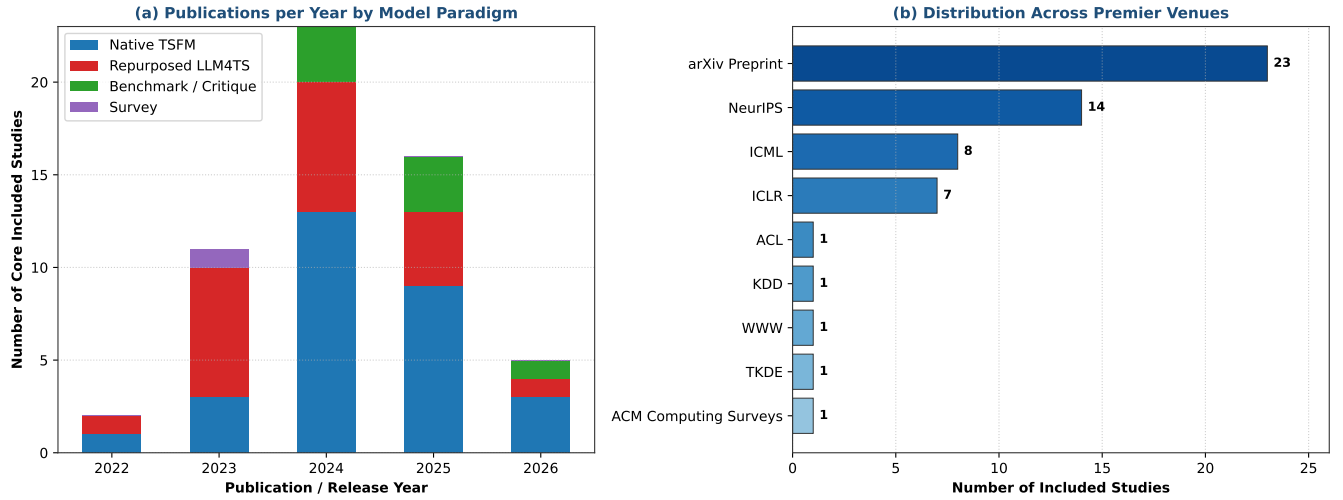


Fig. 4. Meta-review statistics: (a) Publications per year broken down by model paradigm (stacked bar chart); (b) Distribution of included studies across premier peer-reviewed venues.

pretraining on synthetic kernel mixtures and diverse physical systems is essential to sustain positive scaling exponents.

7 Open Challenges and Future Horizons

Despite rapid progress in foundation models and LLM adaptations, temporal machine learning remains in a formative phase. In this section, we delineate six critical open research horizons that must be addressed to achieve truly universal temporal intelligence.

7.1 Irregular Sampling and Continuous-Time Dynamics

Existing foundation models (e.g., Chronos, TimesFM, MOIRAI) predominantly assume regularly sampled, synchronous discrete grids. However, real-world physical telemetry—such as ICU medical records, IoT sensor dropouts, and financial trade ticks—arrives at asynchronous, irregular time intervals. Emerging approaches such as FlowState [41]

leverage continuous flow matching and continuous-time neural state-space models to achieve sampling-rate equivariance. Future foundation architectures must natively encode irregular time intervals (Δt) into latent manifolds rather than relying on crude imputation or resampling heuristics.

7.2 Prior-Data Fitted Bayesian Networks

Standard foundation models require thousands of gradient optimization steps during fine-tuning. A compelling alternative is demonstrated by TabPFN-TS [48], which adapts Prior-Data Fitted Networks to time series. By training a transformer exclusively on synthetic prior stochastic processes (e.g., Gaussian processes, Ornstein-Uhlenbeck, and Markov switching models), the network learns to perform Bayesian posterior inference in a single forward pass without parameter updates. Expanding synthetic prior spaces to encompass complex non-linear dynamics represents a fertile research direction.

7.3 Extreme Long-Horizon Context and Serial Scaling

In natural language, modern models handle context windows exceeding 1M tokens. In temporal analysis, historical series often span millions of high-frequency timestamps. Capturing decadal climate shifts, annual retail cycles, and second-level power grid vibrations within a single context window demands extreme context scaling. Architectural innovations such as long-context state-space models (Mamba), hierarchical linear attention, and serial scaling (as explored in Timer-XL [16] and Timer-S1 [18]) will be essential to eliminate context truncation.

7.4 Unified Multimodal Perception

Real-world decision-making rarely relies on isolated scalar numbers. An industrial operator monitors pressure gauges alongside diagnostic maintenance logs; a macroeconomist reads financial reports while analyzing stock tickers. Future foundation models must advance beyond ad-hoc text reprogramming to train native multi-modal architectures from scratch—treating numerical time series, textual reports, spatial satellite imagery, and audio vibrations as co-equal perceptual modalities.

7.5 Calibrated Uncertainty under Distribution Shift

In safety-critical applications (such as healthcare and grid management), an uncalibrated point forecast is hazardous. As demonstrated by recent reliability audits [26], current zero-shot models frequently underestimate predictive variance when confronted with unprecedented macroeconomic shocks or extreme weather events. Developing conformal prediction techniques, robust Bayesian heads, and flow-based generative sampling that maintain rigorous coverage guarantees under out-of-distribution shifts is imperative.

7.6 Edge Deployment and Hardware Quantization

While research models scale to billions of parameters, industrial IoT environments require deployment on low-power microcontrollers, edge gateways, and automotive Electronic Control Units (ECUs). Adapting post-training quantization (INT8/INT4), model pruning, and specialized kernel architectures (following IBM’s TTM philosophy [13]) will be necessary to bridge the gap between academic benchmarks and enterprise production lines.

8 Conclusion

In this survey, conducted under the PRISMA 2020 systematic review protocol, we have synthesized the foundational developments at the intersection of Large Language Models and Time Series Foundation Models from 2021 through 2026. We formalized mathematical definitions of temporal sequences, subseries patching, and quantization; introduced a unified five-dimensional taxonomy; and compared 41 core benchmark and foundation models across architecture, tokenization, pre-training corpora, and uncertainty formulations.

Our critical synthesis highlights a decisive architectural demarcation:

- Native TSFMs (such as Chronos, TimesFM, MOIRAI, Timer, and TTM) demonstrate superior parameter efficiency, faster inference latency, and stronger zero-shot numeric forecasting accuracy by training directly on temporal dynamics without linguistic mismatch.
- Repurposed LLMs (such as Time-LLM and multimodal reasoning frameworks) excel not at raw numeric extrapolation, but as cognitive orchestrators—grounding numerical time series within textual narratives, metadata, and decision-support systems.

Looking ahead, the frontier of temporal intelligence lies in resolving benchmark contamination, achieving continuous-time sampling equivariance, calibrating uncertainty under non-stationary shifts, and delivering ultra-efficient edge deployment. We hope this survey provides a rigorous compass for researchers and practitioners building the next generation of universal temporal models.

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